CHAPTER – 05

LIST

Introduction to Lists in Python

A **list** is a built-in data type in Python used to **store multiple items** in a **single variable**. It can store elements of different data types such as integers, floats, strings, etc

Key Features of Lists:

1. **Ordered**:
   Items in a list maintain their order. When you add a new item, it is placed at the end.
2. **Changeable (Mutable)**:
   You can change, add, or remove items after the list has been created.
3. **Allows Duplicates**:
   Lists can have duplicate values.
4. **Indexed**:
   Just like strings, list items can be accessed using index numbers (starting from 0).

Syntax of List :

```
Code : my_list = ["a", "b", "c"]

       print(my_list)
```

Mixed Data Types in a List :

```
code :   mixed_list = [10, 3.14, "Hello", True]

     print(mixed_list)
```

List Indexing in Python

Just like strings, **lists are indexed** — each element has a position number, starting from 0.

Indexing Syntax:

```
code : list_name[index]
```

example index : fruits = ["apple", "banana", "cherry", "mango"]

Element	Index	Negative Index
"apple"	0	-4
"banana"	1	-3
"cherry"	2	-2
"mango"	3	-1

Accessing Elements:

```
print(fruits[0])    # apple
print(fruits[2])    # cherry
print(fruits[-1])   # mango (last element)
print(fruits[-3])   # banana
```

Index Error Example:

```
print(fruits[4])    #  IndexError: list index out of range
```

LIST METHOD :

consider the following list :

Method	Description	Example Code
append(x)	**Adds an element x at the end of the list**	**fruits.append("grape")**
clear()	**Removes all elements from the list**	**fruits.clear()**
copy()	**Returns a shallow copy of the list**	**new_list = fruits.copy()**
count(x)	**Returns the number of occurrences of value x in the list**	**fruits.count("apple")**

index(x)	**Returns the index of the first occurrence of value x**	**fruits.index("banana")**
insert(i, x)	**Inserts element x at position i**	**fruits.insert(1, "orange")**
pop(i)	**Removes and returns the item at index i (last item by default)**	**fruits.pop() or fruits.pop(2)**
remove(x)	**Removes the first occurrence of value x from the list**	**fruits.remove("apple")**
reverse()	**Reverses the order of the list**	**fruits.reverse()**
sort()	**Sorts the list in ascending order (or descending with reverse=True)**	**fruits.sort() or fruits.sort(reverse=True)**
len(list)	**Returns the number of items in the list**	**len(fruits)**
type(obj)	**Returns the type of the object**	**type(fruits)**
Slicing	**Used to access a part of the list using [start:end:step]**	**fruits[1:3] or fruits[::-1]**

Code :

```
fruits = ["apple", "banana", "cherry"]

# append
fruits.append("mango")        # ['apple', 'banana', 'cherry', 'mango']

# insert
fruits.insert(1, "orange")    # ['apple', 'orange', 'banana', 'cherry', 'mango']

# remove
fruits.remove("banana")       # ['apple', 'orange', 'cherry', 'mango']
```

 # pop

```
fruits.pop()                # removes 'mango'
```

 # reverse

```
fruits.reverse()            # ['cherry', 'orange', 'apple']
```

 # sort

```
fruits.sort()               # ['apple', 'cherry', 'orange']
```

 # count

```
print(fruits.count("apple"))    # 1
```

 # index

```
print(fruits.index("orange"))   # 2
```

 # copy

```
new_list = fruits.copy()
```

 # clear

```
fruits.clear()              # []
```

Creating Lists in Python

In Python, a list is a collection which can store multiple items. Lists can be created in the following 3 ways:

(i) Homogeneous List : All elements in the list are of the same data type (e.g., all integers).

```
numbers = [1, 2, 3, 4, 5, 6, 7]

print(numbers)
```

(ii) Heterogeneous List

List can contain **different data types** — integers, floats, strings, booleans, etc.

```
mixed = [1, "a", 2.5, True]

print(mixed)
```

(iii) Nested List

A list inside another list (multi-dimensional list).

```
matrix = [[1, 2, 3], [4, 5, 6], [7, 8, 9]]

print(matrix)
```

Accessing List Items

You can access individual items in a list using their index number indexing start from 0**:**

Example 1 – Simple List:

numbers = [1, 2, 3, 4, 5, 6]

print(numbers[1]) # Output: 2 (second element)

Example 2 – Nested List (2D List):

matrix = [[1, 2, 3], [4, 5, 6], [7, 8, 9]]

print(matrix[1][2]) # Output: 6 (3rd item in 2nd list)

3. Modifying List Elements

Lists in Python are **mutable**, meaning you can **change** their values.

Example:

numbers = [1, 2, 3, 4, 5, 6]

numbers[3] = 55

print(numbers)

Output:[1, 2, 3, 55, 5, 6]

4. Adding Elements to a List

You can add elements using **3 main methods**:

Method	**Description**	**Example**
append(x)	Adds x at the end of the list	numbers.append(10)
insert(i, x)	Inserts x at position i	numbers.insert(2, 99)
extend(list2)	Adds all items from list2 to the list	numbers.extend([7, 8, 9])

Code Example:

numbers = [1, 2, 3]

numbers.append(4) # [1, 2, 3, 4]

numbers.insert(1, 99) # [1, 99, 2, 3, 4]

numbers.extend([5, 6]) # [1, 99, 2, 3, 4, 5, 6]

print(numbers)

5. Removing Elements from a List in Python

You can remove elements from a list using several built-in methods depending on what you want to do.

Method	**Use Case**	**Example**
remove(x)	Remove by **value**	list.remove("apple")
pop(i)	Remove by **index**, or last if not given	list.pop(2) or list.pop()
clear()	Remove **all elements**	list.clear()
del list[i]	Delete element at index i	del list[0]
del list	Delete the whole list	del list

PRACTICE SET

Python Practice Questions — List Operations

Instructions:

Write a Python program to perform the following list operations.

Question 1:

Create a list of at least 10 elements.
Access and print the **3rd and 4th** position elements from the list.

Question 2:

Modify the elements at the **2nd and 9th** positions of the list.
Print the list after modification.

Question 3:

Add a new element at the **end** of the list using a suitable method.
Also, **insert** an element at a **specific position** (e.g., 5th position).

Question 4:

Remove an element from the list at a **specified index** (e.g., 6th position).
Print the list after removal.

Question 5:

Slice the list into multiple parts (e.g., part1 = first 4 elements, part2 = next 4).
Access and print **any two parts** of the list.

Question 6:

Sort the list in both:

- **Ascending order**
- **Descending order**
  Print both results.